Zhipeng Jiang
New Jersey Institute of Technology
4^{th} Year PhD Candidate in Mechanical Engineering & Computer Science
Newark College of Engineering
Department of Mechanical and Industrial Engineering Department
University Heights, Suite 204
Newark, New Jersey 07102
937-823-5986
zj28@njit.edu | zjiang452@gatech.edu

Education

Georgia Tech Institute of Technology – Master of Science in Computer Science, August 2025-Now
- Program: Online Master of Science in Computer Science (GT OMSCS)
- Course works:
  - **Artificial Intelligence (Knowledge-Based AI)**,
  - **Software Development (Software Architecture and Design)**
  - **Agentic AI Seminar (Agentic AI Essentials)**
  - **ML for Sensor-based Human Activities Seminar (ML for Sensor-based HAR)**
- Target Specialization: **Interactive Intelligence**

New Jersey Institute of Technology- Doctor of Mechanical Engineering, August 2022-Now
  - Dissertation Advisor: Lin Dong
  - Mechanical Engineering PhD Candidate (Current GPA 3.75/4.0)
- Research focus: **Smart Flexible Sensor-based Human Robotic Interaction with VR & ML**

Mississippi State University– Master of Aerospace Engineering, August 2020 - December 2023
- Mechanical Engineering (GPA 3.75/4.0)
- Research focus: **Dynamics, Acoustics, and Vibration Control System**

University of Dayton– Bachelor & Master of Mechanical Engineering, August 2014 – May 2020
- Mechanical Engineering (GPA 3.54/4.0)
- Undergraduate Honors Thesis: **Design of a Body Powered Prosthetic Arm**

Positions
- **Teaching / Research Assistant**, New Jersey Institute of Technology, Newark, NJ, 2022-now
  - Developed multi-biosignals (EMG/ECG/EEG) acquisition pipelines with ML methods (RCNN).
- **International Peer Academic Coach**, University of Dayton, OH, 2018
- **Mechanical Engineering Co-op**, Exsplendid Inc. Dayton, OH, 2017.

Awards

- **Z. Jiang**, N. Kiwanian, (2025, December 12) Limb*VR* – A VR-based rehabilitation platform combining wearable resistive strain and flexible EMG sensing with a Unity-controlled virtual hand to enable real-time prosthetic training and neuromotor control studies. ***2025 NJ New Business Model Competition***. Drexel University, Philadelphia, PA, US. (**Won 2nd Place, selected joining 2026 Hult Prize Competition National Round)**.

- **Z. Jiang**, N. Zhang, N. Kiwanian, (2025, October -2026, January) *Wetrophia* – An AI-Agent /GYN triage assistant that chats with patients, gathers essential symptoms, evaluates urgency, routes them to the right doctors, and produces a clear, reference-backed appointment summary using RAG System. ***RU Health-Hackathon 2025***, Rutgers University, New Brunswick, NJ, US. (**Honorable Mentioned Award, selected for Rutgers 2025 Women's Health Accelerator Program**)
  - Wetrophia | Github

- **Z. Jiang**, T. Li, (2025, September 12) *ReBloom* – Built an AI-powered AR wellness coach using wearable EMG sensors and Python-based real-time signal analysis to guide personalized postpartum recovery, ***2025 HopHacks***. John Hopkins University. Baltimore, US. (**Won Best AI & AR for Maternal Wellness)**
  - ReBloom | Devpost

- **Z. Jiang**, Z. Li, Z. Li (2025, May 31) *RPG Craft* – An Agentic AI-powered and generatively scripted game with LangFlow-based multi-AI-Agent System. ***2025 Hacking Agent NYC Hackathon***. New York, US. (**Won Best Agentic AI Design Award)**
  - 🏆 Hacking Agents Hackathon NYC: Winners Showcase 🗽 | LinkedIn

- **Z. Jiang**, N. Kiwanian, X. Li (2025, April 27). *Cybathlon VR – Feeling Empowered: Driving Innovation in Rehabilitation Science with an interactive VR rehabilitation system that fuses wearable resistive strain and flexible EMG sensing into a Unity-controlled virtual hand for real-time prosthetic training and motor-control research*. ***2025 Dragon Hackathon***. Drexel University, Philadelphia, PA, US. (**Won 1st Place on Medical Track)**
  - Cybathlon VR | Devpost

- **Z. Jiang**, H. Fang, T. Li, Z. Yazdani, J. Dicker, R. Tiliouine, S. Lu, J. Xie, T. Nguyen, A. Mukherjee, S. Hu. (2024, October 26) *Heartup,* an AI-powered cardiac VR-based digital twin that combines Bi-LSTM with attention and MLP models to analyze ECG data, predict long-term health risks, and support patient–provider decision-making. ***RU Health-Hackathon 2024***, Rutgers University, New Brunswick, NJ, US. (**Won 4th Place with Honorable Mention Award)**
  - HeartUp | Github

- **Z. Jiang**. (2024, April 10). *Smart Human-machine Interface*, a self-powered, breathable piezoelectric human–machine interface that translates bicep muscle signals into Morse Code for gesture recognition and robotic prosthetic control. **2024 NJIT Dana Knox Research Showcase**. Newark, NJ, US. (**Won 1st Place)**
  - NJIT's Dana Knox Research Showcase Celebrates Future Visionary Leaders

Research Works in Progress

- Currently working on attaching smart flexible sensors to robotics with VR & Agentic AI to interact with humans to intelligently improve our daily life.

Conference Presentations

- S. H. Kwon, N. Ratnakumar, **Z. Jiang**, Z. You, X. Zhou, L. Dong. (2023, October 9- 11). A self-powered flexible wearable sensor for activity recognition. IEEE-EMBS International Conference on Body Sensor Networks: Sensor and Systems for Digital Health (IEEE BSN 2023). Boston, MA, US.

Publications

- Wang, Y., **Jiang, Z.**, Kwon, S., Ibrahim, M., Dang, A., Dong L., (2025). Flexible Sensor-based HMIs with AI for Medical Robotics" *Advanced Robotics Research*, p.202500027
- Zhang,C., Kwon, S.H., Huerta, A., **Jiang, Z.**, Sun, M., Dong, L., (2025) Self-Polarized Piezoelectric Hydrogels with Flexible-Rigid Networks for Ultrasensitive Multifunctional Sensors, *ACS Materials Letters,* vol. 7, p.2607-2616
- Zhang, C., **Jiang, Z**., Sun, M., Augustin-Lawson, R., Kwon, S. H., & Dong, L. (2025). Nature-Inspired Helical Piezoelectric Hydrogels for Energy Harvesting and Self-Powered Human-Machine Interfaces. *Nano Energy*, 110755.
- **Jiang, Z**., Zhang, C., Kwon, S. H., & Dong, L. (2024). Self-Powered, Soft and Breathable Human–Machine Interface Based on Piezoelectric Sensors. *Advanced Sensor Research*, *3*(12), 2400086.
- Kwon, S. H., Zhang, C., **Jiang, Z**., & Dong, L. (2024). Textured nanofibers inspired by nature for harvesting biomechanical energy and sensing biophysiological signals. *Nano Energy*, *122*, 109334.
- Zhang, X., Li, L., Pu, L., Yang, J., Wang, Z., Fu, R., & **Jiang, Z**. (2024, February). Deep Learning-based Malicious Energy Attack Detection in Sustainable IoT Network. In *2024 International Conference on Computing, Networking and Communications (ICNC)* (pp. 417-422). IEEE.
- Lu, Z., Zhang, C., Kwon, S. H., **Jiang, Z**., & Dong, L. (2023). Flexible Hybrid Piezoelectric-Electrostatic Device for Energy Harvesting and Sensing Applications. *Advanced Materials Interfaces*, *10*(8), 2202173.
- Zamiela, C., **Jiang, Z**., Stokes, R., Tian, Z., Netchaev, A., Dickerson, C., ... & Bian, L. (2023). Deep multi-modal U-Net fusion methodology of thermal and ultrasonic images for porosity detection in additive manufacturing. *Journal of Manufacturing Science and Engineering*, *145*(6), 061009.
- Al Mamun, A., Bappy, M. M., Mudiyanselage, A. S., Li, J., **Jiang, Z**, ... & Tian, W. (2023). Multi-channel sensor fusion for real-time bearing fault diagnosis by frequency-domain multilinear principal component analysis. *The International Journal of Advanced Manufacturing Technology*, *124*(3), 1321-1334.
- Righi, H., Li, T., **Jiang, Z**., Li, J. (2022). Numerical Study of Ultrasonic Guided Waves in a Composite-Plastic Lap Joint. *IFAC-PapersOnLine*, *55*(27), 353-357.
- Zamiela, C., Tian, W., Bian, **L., Zhipeng**, J., (2021). Deep fusion methodology of infrared and ultrasonic data for porosity detection in additive manufacturing. In *IIE Annual Conference. Proceedings* (pp. 229-234). Institute of Industrial and Systems Engineers (IISE).